

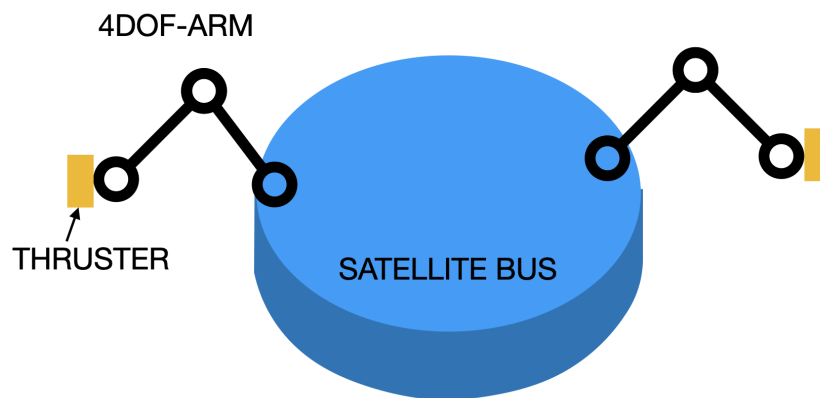
Take-Home (≤ 5 hours): Dual-Arm + End-Effector Thrusters Simulation and Control

Build a physics-engine sim of a free-flying vehicle (satellite) in 0 gravity with two 4-DoF arms rigidly mounted to it. Each arm has a thruster at its end-effector.

What we need

1. Sim

- Vehicle is not fixed to the world (free-floating).
- Two arms, 4 joints each (your choice of joint types, kinematics).
- Two end-effector thrusters (one per arm) that apply force at the EE when enabled. Thrust direction up to your choice.
- Feel free to choose mass, actuator gains, link length and so on, justify your choices.



Reference Sim Setup

2. Teleop

- Keyboard controls to move all 8 joints (increment/decrement is fine).
- Toggle left thruster on/off, right thruster on/off (optional throttle 0–1).
- Reset simulation.

3. Control

Integrate a software component to move the satellite from its current position to a user provided new pose. You may use the control strategy or methodology of your choice. Please justify your choices.

Include a couple plots (base pose/vel + thruster commands) and a short video/GIF showing control + thrust causing vehicle motion.

4. **Configurable parameters**

Make it easy to tweak without code edits:

- vehicle mass/inertia (or swap asset)
- link masses (or per-arm scaling)
- max thrust per thruster

Short write-up (2–4 paragraphs)

Describe your choices and how you selected each value. Explain the high-level control approach (planning + feedback + thrust allocation), and how you'd handle limits (joint/thrust/rates). Explain how you would scale this approach to a hypothetical demonstration mission happening next year.

Deliverables

Repo/zip + README (run steps + key bindings) + logs + plots + video/GIF + write-up.